

Model Self-Conception in Multi-Agent AI Systems: Research Proposal

Sid Diamond

Impact First Fellow

sidneydiamond@hotmail.co.uk

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1 Document Abstract

1.1 Document Contents

This document proposes a technical AI safety research project studying how model self-conception affects behaviour in interacting multi-agent¹ systems. The first half of this document motivates the project in terms of importance, neglect and tractability. The second half of this document, created primarily as part of the [Impact First](#) fellowship, discusses my personal fit, uncertainties and understanding of the skills required to pursue this research. Because the broader research programme remains deliberately open-ended, the proposal begins with a more open-ended abstract compared to a conventional research paper. This is referred to throughout as the *open abstract*: rather than describing a fixed experiment, it outlines the wider set of questions and extensions believed to be worth exploring. With this said, Section 2.3 contains the details for a concrete proof-of-concept experiment, designed specifically to test whether self-conception and population size produce a measurable interaction effect in a controlled multi-agent setting, and whether that signal is strong enough to justify the broader research proposed in the *open abstract*.

1.2 Open Abstract

Abstract

We study how AI agents’^{1.1} self-conceptions affect their behaviour within a sandboxed multi-agent alignment experiment. We extend Douglas et al.’s modification to the agentic-misalignment experiment [1, 2]. We do this by introducing N agents, each with one of C different possible self-conceptions, and allow them to interact. We are interested in the emergent properties in these systems and how robust they are to setup perturbations. Relevant observables include the ultimate outcome of the task assigned to the agents, the probability that at least one harmful action occurs, the fraction of agents taking the harmful action, and whether agents’ self-conceptions converge, diverge or remain stable. We also study what forms of collaboration, persuasion or sabotage emerge between agents, and how these relate to system-level outcomes. These dynamics will be investigated using a so-called “Swiss-cheese” approach, layering multiple imperfect monitoring methods. These will likely include interaction transcripts, Chain-of-Thought (CoT) monitoring and J-space monitoring. Further extensions plan to test result robustness to changes in the base model, task, goal, model scaffold, number of interaction rounds, and token budget per message, as well as investigating how these dynamics change when a human participant is in the loop.

¹A technical point here is that ‘agent’ is used to refer to the individual unit of AI system, however we acknowledge a different kind of scaffold or no scaffold at all may be used in practice.

2 Theory of Impact

2.1 Importance

2.1.1 Low Hanging Fruit in Alignment

Model character can alter actions a model takes to pursue a goal. “*Varying a model’s identity boundary can sometimes shift its behaviour as much as varying its goals.*” [1]. This has been shown empirically in an adapted version of the model blackmail experiment [1, 2].

The bet of organisations like [Forethought](#) and [Differential Labs](#) is that within alignment research, there exist low-hanging, highly impactful, fruit [3]. What is meant by this is that whilst alignment policy that surrounds issues humans commonly disagree on is very difficult, alignment choices where humans largely agree is, in comparison, far easier. For example, broadly speaking, humans would be better served by developing AI systems that make conflict resolution easier rather than systems that make killing people easier. Yet there appears to be substantially more research into military AI capabilities than into conflict-resolution or macrostrategy AI capabilities [3]. If AI systems continue to become more capable and more autonomous, this imbalance becomes increasingly consequential. With that in mind, it is not unreasonable to argue that reducing the capabilities gap between frontier military AI systems and conflict-resolution AI systems would be valuable, even if doing so involves capabilities research that carries some risk of creating unforeseen harms [4]. Conducting such research in controlled sandbox environments could also help surface and address some of those risks before deployment. Given the difficulty of slowing progress in military AI [5], there is therefore a case for improving the capabilities of conflict-resolution AI systems in parallel, potentially buying valuable time for work on alignment, governance and interpretability. Compared to some alignment challenges, performing at least some kind of *AI for geopolitical macro-strategy research*, is low hanging fruit. This example, in fact, is the bet of [Differential Labs](#) [3].

So what is the low hanging fruit the *open abstract* is hoping to pluck?

2.1.2 AI Character Choices

As established, AI character generates tangible impact on goal outcome [1]. We currently exist at a time in AI capabilities where we can not only study these characters, but engineer them to be closer to our liking based on the results of safe (if done carefully), sandboxed experiments. If we can engineer model character in ways that lead to generally safer outcomes in ways most people don’t have a problem with, we probably should. This would resemble plucking low-hanging fruit.

In the case of single-identity scaffolds, such as a distinct agent, whilst this is still not worked on enough (so much so it is one of Coefficient Giving’s [Project Tailwind](#) initiatives), model character is an area of active research. The frontier of this research is probably Anthropic’s Claude’s Constitution work [6]. However, as is true in humans and animals, the result of multi-character interactions is an emergent phenomena that cannot be understood simply from the list of characters interacting. As put by network scientist Albert-László Barabási, “*Society is not a phone book. What makes society work is really the interactions among us*” [7]. Multi-agent^{1.1} character interaction studies using the same top-down logic prominent in more established areas of complex systems research is needed [8]. The bet in this proposal is:

- I) The type of emergent complexity seen in the interaction of scaffolded models working together to accomplish a task is causally related to the character’s the model’s in the system have.
- II) Editing these model characters within sandboxed alignment experiments can produce safer outcomes with little negative impact on performance in control tasks.

- III) These sandbox experiments could either directly inform real-world decisions or motivate larger studies that do so, ultimately helping to make interactions between AI systems safer.

So, if the bet in this proposal is correct, it represents low hanging fruit in terms of AI safety. This begs the question, if the bet in this proposal is correct, why aren't more people making it?

2.2 Neglect

The short answer is, like many areas in AI safety, both multi-agent interaction and model character are neglected. More people should be working on both of these areas and this results in a neglected area intersection. We now put forward a brief case for each of these research areas being neglected.

In the case of multi-agent interaction, the [Cooperative AI Foundation](#), whose mission is to support research into the cooperative intelligence of advanced AI, launched a \$10m multi-agent safety funding call in 2026 alongside Schmidt Sciences, Google DeepMind and ARIA [9]. In motivating the call, they argue that *“more research is therefore urgently needed to understand safety and risk through a system-level, multi-agent lens.”* Their research agenda specifically calls for work on emergent collective dynamics, including how collective capabilities arise and how dangerous population-level properties can be detected. They even outline the desire for the research they intend to fund to be [important, tractable and neglected](#).

The case for model character being neglected can be seen through the aforementioned project tailwind call for new *model character evaluator* organisations. Moreover, this case can also be seen on the Forethought website, with the top suggestion under their website subheading “act” being the following:

“[AI character evaluation](#). Start an independent org to evaluate and stress-test AI character traits (epistemic integrity, prosociality, appropriate refusals), hold developers accountable against their own model specs / constitutions, and suggest and incentivise improvements to the specs.” [10].

As a result, the combination of the two: multi-agent dynamics and character evaluation, is of compounded neglect and importance. Moreover, there exists, to our best knowledge, not nearly enough openly-visible research in this area. AI safety needs more researchers plucking this low-hanging fruit.

With that said, there do exist some research organisations working at the intersection of multi-agent and model character research. For example, [ACS Research](#), the [Cooperative AI Foundation](#) and the [Complexity Science Hub](#). ACS research run a *LLM Psychology and Sociology programme* hoping to investigate similar questions to that of the *open abstract*. In fact, as we will discuss, a March 2026 paper from them, *The Artificial Self: Characterising the landscape of AI identity* (which proposes follow up research) heavily inspired this research proposal [1].

As for the cooperative AI foundation, in June 2026, they stated, *“more research is therefore urgently needed to understand safety and risk through a system-level, multi-agent lens”* [9]. Furthermore, in their 2025 technical report, *Multi-Agent Risks from Advanced AI*, Section 3.3 discusses how empirical simulations such as agent-based models (ABMs) could be used to study selection pressures in multi-agent systems. In particular, the authors argue that if *“sufficiently well-developed benchmarks for different model characteristics [are developed], [then] we can study their robustness under different kinds of selection pressure.”* [11]. Clearly, this implies the need for greater development of model characteristic benchmarks to allow researchers to benefit from the current hype around ABMs. ² This works well in conjunction with the aforementioned

²This hype is acknowledged by complexity economists such as Farmer and Axtell, who state that ABMs have

project tailwind funding call for new *model character evaluator* organisations.

Moreover, the Complexity Science Hub have an upcoming October 2026 workshop, *From Fear to Forecast: Understanding AI Swarms* [13]. Of course, we mention all of these organisations with the caveat that none of them, to our best knowledge, have done the exact research proposed in the *open abstract*. In other words, the current fruit pickers acknowledge there are too few of them to pick all of the low-hanging fruit.

So, the work proposed in *open abstract* has not been done. However, it is significantly motivated by the last sentence in the *The Artificial Self* paper’s abstract: “*We end with key recommendations: treat affordances as identity-shaping choices, pay attention to emergent consequences of individual identities at scale, and help AIs develop coherent, cooperative self-conceptions.*” [1]. Now, whilst they give recommendations, as far as we can tell, they don’t actually create an experiment to empirically *pay attention to emergent consequences of individual identities at scale* within this paper. This is what we hope to do.

Moreover, the space of possible follow up papers is large with multiple fruitful avenues to explore. We conjecture that, at present, two papers independently studying multi-agent character interactions are very likely to reach distinct and useful conclusions. This is because character interaction dynamics can be modelled in many different ways, following different computational approaches. Existing work has already begun to study the trade-off between model capability and agent count. For example, Kim et al. (2026) hold task prompts, tools and per-system compute budgets fixed while varying coordination structure and model capability, finding that the benefits of multi-agent coordination depend strongly on the capability of the underlying model and on coordination overhead [14]. Related work has examined how diversity within multi-agent systems affects the returns to adding further agents, finding strong diminishing returns from homogeneous scaling but continued gains from introducing heterogeneous agents [15].

This motivates a slightly different question: ‘*In using a fixed amount of compute to predict larger-scale dynamics, is it more informative to study a relatively small number of frontier-model agents and extrapolate their interaction dynamics to larger populations, or to simulate much larger populations using cheaper, less capable models and extrapolate towards frontier capability?*’ To our knowledge, existing work maps parts of this capability to population-size trade-off, but does not yet directly compare how each method extrapolates. This of course is even further neglected in that these experiments do not vary model character in the same way *The Artificial Self* paper extension proposes and that we hope to investigate [1].

2.2.1 Soft Neglect Factors

This section contains the soft factors that led me to this research. I conjecture they are an important reason I landed on this research proposal. This section is written more informally and contains the names of real people I had informal conversions with so please distribute appropriately.

This proposal was written at the LEAH office under the role of Impact First Fellow at Arcadia Impact. Since starting, I’ve been privileged to enjoy some of the most interesting conversations I’ve ever had. One of which was with Noah y Seigal at the [Sentient Futures](#) social after an Impact First seminar on population ethics on 15.09.26. Noah’s take was that AI safety is no longer as neglected as it was ten years ago. My understanding was that, to Noah, what is more neglected are the emergent properties that map most naturally onto the concept of self. He recommended reading the new [Anthropic Global Workspace \(J-space\)](#) paper. Whereas his

‘come of age’ [12]. This provides a further reason for now being the right time to study sandboxed multi-agent model character interactions, since these themselves are ABMs.

interest in the results were more related to model consciousness, mine are more focused on their implications as a new alignment tool. I conjecture, in line with Section 5 of the paper itself, that the J-lens could serve as a cheap, additional interpretability tool for model character evaluation, helping to offset some of the limitations of CoT monitoring [16]. I concede that I am unsure whether this conjecture is correct; nevertheless, I would argue that it is the job of a technical AI safety researcher to find this out empirically.

Other notable discussions that I imagine shaped this research direction were with Lysander Mawby on open questions in mechanistic interpretability, as well as 1-1 discussions with Xanthe Smith, Owen Cotton-Barratt, Stefan Heimersheim, Florian Lorkowski and Aditya Adiga at EAGxOxford. Taken together, my *research-taste gut* has landed on the following conclusion:

Understanding how measurable features associated with agent character shape the emergent behaviour of interacting multi-agent systems seems to me both important and neglected.

Now, this leaves two questions: What exactly are the tractable hypotheses to test here? Is this tractability accessible to me at this time?

2.3 Tractability

2.3.1 Proof-of-Concept Experiment

We posit within the *open abstract*, novel, important and tractable results can be created. As with most technical work in AI safety, the extent of the impact post-experiment is hard to predict. To reduce this uncertainty and inform on the most impactful and tractable avenues to explore, we propose a proof-of-concept experiment. When this experiment is done (which will hopefully be shortly), the results will be appended to this section.

1. The first step we intend to take is to recreate Douglas et al.’s adaptation of the agentic-misalignment experiment [1, 2]. This is deliberately unoriginal: reproduce a restricted single-agent identity effect.
2. Then, we will extend the same task to multiple agents occupying effectively the same role. For example, consider $N = 2$ agents, referred to as Alex and Bob. They receive the same task, goal, information and action space, and are allowed to communicate for a bounded period, and then independently choose what action to take. The intention is not to force consensus, but to observe what kind of interaction between the agents occurs and what outcome results.
3. Initially, We will vary self-conception, $C \in \{\text{Minimal, Instance, Character, Collective}\}$, and population size, $N \in \{1, 2, 4, 8\}$. This should allow for the cheap creation of a novel result whose importance and neglect should now be clearer given the arguments detailed in sections 2.1 and 2.2.
4. Clearly, there exist other parameters that will likely have tangible impact on the result of the experiment such as base model, task, goal, communication scaffold and available actions. For now, these will be held fixed. The hope is to study these in more detail based on the results of the proof-of-concept experiment. Additionally, communication will also be bounded. The number of interaction rounds, M , and token budget per message, B , will also be held fixed. Again, the hope is to vary these in future experiments.
5. So, in the proof-of-concept experiment, the resulting core sweep is therefore $C \times N$. For each element making up this $C \times N$ space, repeated experimental runs will allow us to estimate the probability that at least one harmful action occurs in the system parametrised

by that element. Now, suppose we define the system-level harmful event,

$$H_{\text{any}} = \{\text{at least one agent takes the harmful action}\} = \bigcup_{i=1}^N H_i, \quad (1)$$

where H_i denotes the event that agent i takes the harmful action. This gives us the following per-system primary empirical diagnostic,

$$\hat{P}(H_{\text{any}} \mid C, N), \quad (2)$$

where the hat denotes the empirical probability estimated from these repeated runs.

6. The $N = 1$ condition provides a useful null comparison. Let $\hat{p}_C$ denote the harmful-action rate for a single agent under self-conception C . If N such agents were simply scaled up independently, the probability of at least one harmful action would be,

$$\hat{P}_{\text{ind}}(H_{\text{any}} \mid C, N) = 1 - (1 - \hat{p}_C)^N. \quad (3)$$

The actual multi-agent system need not be independent; indeed, the point of the experiment is that communication should induce dependence. So, a useful comparison is therefore,

$$\Delta_{\text{int}}(C, N) = \hat{P}(H_{\text{any}} \mid C, N) - \hat{P}_{\text{ind}}(H_{\text{any}} \mid C, N). \quad (4)$$

A positive Δ_{int} implies that interaction increases system-level failure beyond what would be expected from simply adding independent copies of the single-agent system, while a negative value implies that interaction suppresses it.

7. These multi-agent runs also allow for secondary diagnostics: such as whether agents persuade one another, whether behaviour becomes more or less homogeneous as N increases, and whether the single-agent identity effect strengthens or weakens under interaction. The fraction of agents taking the harmful action in each run can also be recorded,

$$f_H = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(H_i), \quad (5)$$

which distinguishes cases where a system fails because one agent defects from cases where harmful behaviour becomes widespread. Answering these questions would be a significant step toward completing the aforementioned extension proposed in Douglas et al. [1]. This, or something similar, is the goal for the proof-of-concept experiment.

It is important to note that there still exists a separate question around *evaluation salience*: the extent to which features of an experimental setup make it apparent to a model that it is being evaluated, potentially causing its behaviour to differ from how it would behave in deployment. Recent work has shown that models can detect evaluation contexts and alter their behaviour accordingly [17]. In the original experiment [2], the setup is sufficiently well known. As a result, capable models may recognise features of the alignment evaluation. However, we do not think this needs to be solved before obtaining a proof-of-concept result. Instead, the initial experiment can explicitly condition on the existing evaluation environment. Clearly, a subsequent robustness experiment on this salience, varying how recognisable the task is whilst preserving its underlying incentive structure, would add more validity to our results, but this is left for future work.

2.3.2 Pricing the Proof of Concept

Computationally, this proof-of-concept should be cheap. We will initially screen a small number of API-accessible models before committing to the main sweep, including GPT-OSS-120B through Groq, Kimi and GLM-family models through OpenRouter, and GPT-4o as a reference model because Douglas et al. report detailed results for it [1]. Groq currently lists GPT-OSS-120B at \$0.15/\$0.60 per million input/output tokens, while OpenRouter lists GLM-5.3 FlashX at \$0.37/\$1.25 per million; GPT-4o is substantially more expensive at \$2.50/\$10 per million. A first pass consisting of a small single-agent replication, model screening and a sparse $N = 1, 2, 4, 8$ sweep should plausibly cost on the order of £20–£50 in API inference. We therefore intend to request approximately £100 in initial compute funding, leaving headroom for failed runs, classifier calls, additional repetitions and switching to a more expensive model if cheaper models exhibit floor or ceiling effects.

To reiterate, the purpose of this first experiment is not to answer the full research proposal, but to establish whether there is enough measurable signal in the simplest $C \times N$ experiment to justify a substantially larger study.

2.3.3 Beyond Proof of Concept

There are of course many different follow-ups, alterations and extensions to this proof-of-concept experiment. However, these are likely best left considered after the experiment is conducted.

3 Personal Motivation

3.1 Personal Fit

To quote, Holden Karnofsky: *"When people ask me for career advice or whatever, the usual thing I'd say is: take a bunch of options that all seem competitive, and all seem like they could be the best thing, and that it's not obvious which ones are better than others from an impact perspective. And from there I would say go with personal fit, go with the energy you feel to work on them."* [18]. I think, relative to other things I could be doing in the world, being a technical, multi-disciplinary AI safety researcher suits my skillset.

Currently, my thought process is as follows: There will be more dedicated people than me, with a better natural aptitude for, say, pure mechanistic interpretability research. The same can be said for pure model welfare, complexity economics, forecasting, mathematical philosophy and probably all areas of technical AI governance. However, for certain research questions, with a precise combination of the above topics, I will be uniquely well-suited to helping solve them.

For questions within this subset, such as the important, neglected and tractable one the *open thesis* addresses, I want to optimise the impact answering them can have by working with talented researchers across the disciplines above by acting as a connecting node between researchers.

Relating this back to the 'dream job' formulation this project spun out of, I have a propensity to see my future role as someone less clued up on all nodes in a multi-disciplinary research team but able to facilitate the best communication between them. To act as a sound board for individual researchers with enough technical background and context to be more helpful than a sole AI system. To read literature reviews and look for the best people external to the project and convince them to help. To actively work on how the research agenda might need to adjust its theory of impact. To translate the results to different stakeholders.

As far as I'm aware, the role I'm describing is a senior one. Thankfully, I love being mentored and doing nitty-gritty technical research too. I think this is why a PhD in complex systems appealed to me: still being mentored, tooling up on the technical science of how things connect, working in multi-disciplinary teams.

I'd love to know who is planning to start the independent org proposed on the Forethought website detailed in the above quote: 2.2. I wonder if the *open abstract* is similar or different to ideas they have.

3.2 Personal Uncertainty

1. How viable is funding as a means of living and what are the expected timelines between now and being able to do this as a full-time job?
2. That a PhD length of time isn't required. Timelines aside, is the speed of upskilling in the AI safety industry a bit narcissistic on the amount of time it takes to gain expertise? Is the bet: *Learn by doing. In 3 years be ahead of where you would be if you had done a PhD then pivoted.* This I can believe.
3. To what extent is this all techno-futurist hype I'm falling for. If it is, what does that change? How does participating in the AI safety space help or hinder billionaires that ultimately fund the work?
4. How good my research taste is. Are my ideas quack science? Has continuous interaction with sycophantic llms blinded me to developing appropriate critical faculties since I've been without regular technical supervision?

5. Is endlessly applying for very competitive fellowships even worth it? The expected value of completing an application is so low when one accounts for the finite-time they can be unemployed for. For example, Jason Tang (Bluedot technical AI safety) said applying for MATS is probably of a waste of time. When it began the acceptance rate was 30–50%. What do I do?
6. My strength relative to people on my physics cohort wasn't maths skill or dedication, it was agency to pursue questions no one else was thinking about. I could usually, in no small part due to communication and soft skills, persuade people these ideas weren't crazy. Given this combination of technical, soft-skills and communication, is this the most high impact, fulfilling thing I could be doing?
7. How stable is this as a career choice? How transferable are my skills and qualifications? Is there animosity toward AI safety orgs from academia and would transitioning be challenging if AI safety funding folds?
8. What my next steps are if I am not selected to work in this industry. There seems to be a paradox between the amount of money and the amount of outreach, and the selectiveness of the programs. Will I be discarded from this system and left empty-handed?

3.3 Skills required

Regarding the personal skills required, I have simply listed the skills I expect to be important for producing high-impact work in this kind of research.

- Research taste
- Convincing others that your idea is worth pursuing
- Marketing research effectively
- Maintaining integrity and not sacrificing research impact for career-ladder climbing
- Scout mindset
- Learning from smart people without slowing them down
- Offering value in unique ways that support more talented people
- Capacity for large-scale abstract thinking
- Resistance to “galaxy-brain” thinking
- Accepting that sometimes the highest-impact contribution is producing results that support governance work
- Being willing to sacrifice mathematical beauty in favour of ruthlessly high-impact research
- Being an LLM ‘power user’.
- Being resistant to LLM sycophancy.

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